

Multistep Spatial and Coordinate Problems

Answer Key

Grade 8 / Pre-Algebra

Quick Answers

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| 1. 17 cm | cm, — |
| 2. 10 | 9. the length of the ramp = 3.9 m, the total edging = 9 m |
| 3. 16 cm | 10. the length of AB = 5, the length of BC = 5, the length of AC = 7.07, — |
| 4. the diagonal path = 50 m, the metres saved = 20 m | 11. the diagonal across the surface = 26.93 m, the length of the rope = 27 m |
| 5. the length of PQ = 6, the length of QR = 8, the length of PR = 10, the perimeter = 24 | 12. the diagonal support = 13 m, the perimeter of the fence = 34 m, the total length of fence and support = 47 m, — |
| 6. the diagonal of the base = 15 cm, the diagonal of the box = 17 cm | |
| 7. both solutions of the equation = -7, 7, — | |
| 8. the longest straight line inside the box = 26.57 | |
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What is verified

8 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 7, 8, 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra

8.G.B.7 – *problems 1, 3, 4, 6, 7, 8, 9, 11*

8.G.B.8 – *problems 2, 5, 10, 12*

Difficulty 1–5 of 5 across 27 tagged checks.

- 1.** Legs 8 cm and 15 cm. Find the hypotenuse.

Solution: Both given sides are legs, so add the squares: $c = \sqrt{8^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289}$. $d = 10$

- 2.** Distance between (1, 2) and (7, 10).

Solution: The gaps are $7 - 1 = 6$ across and $10 - 2 = 8$ up. Those are the legs of a right triangle whose hypotenuse is the segment itself: $d = \sqrt{6^2 + 8^2} = \sqrt{100}$. $d = 10$

- 3.** Hypotenuse 20 cm, one leg 12 cm. Find the other leg.

Solution: The longest side is given, so the missing side is a leg and the squares are subtracted: $\sqrt{20^2 - 12^2} = \sqrt{400 - 144} = \sqrt{256}$. 16 cm

4. Field 30 m by 40 m. Find the diagonal, and the saving over walking two sides.

Solution: The two sides meet at a right angle, so they are the legs and the diagonal is the hypotenuse: $\sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50$. Rosa walks $30 + 40 = 70$ m, so the saving is $70 - 50$.

$\text{diagonal} = 50 \text{ m} \quad \text{saving} = 20 \text{ m}$

5. Triangle $P(0, 0)$, $Q(6, 0)$, $R(6, 8)$: find all three sides and the perimeter.

Solution: PQ is horizontal, so $PQ = 6$. QR is vertical, so $QR = 8$. For the slanted side, $PR = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$. Adding: $6 + 8 + 10$.

$PQ = 6 \quad QR = 8 \quad PR = 10 \quad \text{perimeter} = 24$

6. Box: base 12 cm by 9 cm, height 8 cm. Find d and D .

Solution: Stage one is the base rectangle, whose sides are the legs: $d = \sqrt{12^2 + 9^2} = \sqrt{144 + 81} = \sqrt{225} = 15$. Stage two uses that answer as a leg alongside the height: $D = \sqrt{15^2 + 8^2} = \sqrt{225 + 64} = \sqrt{289}$.

$d = 15 \text{ cm} \quad D = 17 \text{ cm}$

7. Solve $x^2 + 576 = 625$ and say which root is the leg.

Solution: Isolate the square, then take both roots:

$$x^2 = 625 - 576 = 49, \quad x = \pm 7$$

$x = -7 \quad \text{or} \quad x = 7$

Reason (instructor-judged). A correct response chooses 7 and explains that squaring hides the sign, so the equation has two solutions, but the side of a triangle is a length and cannot be negative. A chosen number with no reason earns no credit.

8. Pencil box 20 cm by 15 cm by 9 cm. Longest line inside, and will a 26 cm pencil fit?

Solution: The longest straight line inside a box is its space diagonal, which takes all three measurements at once:

$$D = \sqrt{20^2 + 15^2 + 9^2} = \sqrt{400 + 225 + 81} = \sqrt{706} = 26.5707 \dots$$

26.57 cm

Decision (instructor-judged). A correct response says the pencil fits: the longest line available inside the box is about 26.57 cm and the pencil is 26 cm, so it goes in corner to corner even though it is longer than the 20 cm edge. Accept any wording that compares the pencil with the space diagonal rather than with one edge.

- 9.** Ramp rising 1.5 m over a run of 3.6 m. Find the ramp and the total edging.

Solution: The rise and the run are the legs, so the sloping ramp is the hypotenuse: $\sqrt{1.5^2 + 3.6^2} = \sqrt{2.25 + 12.96} = \sqrt{15.21} = 3.9$. The edging runs along all three sides: $1.5 + 3.6 + 3.9$.

ramp = 3.9 m	edging = 9 m
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- 10.** $A(-1, 2)$, $B(3, 5)$, $C(6, 1)$: find the three sides and show the triangle is isosceles.

Solution:

$$AB = \sqrt{4^2 + 3^2} = \sqrt{25}, \quad BC = \sqrt{3^2 + 4^2} = \sqrt{25}, \quad AC = \sqrt{7^2 + 1^2} = \sqrt{50}$$

The first two are exactly equal; the third is $\sqrt{50} = 7.0710\dots$, which rounds to 7.07.

$AB = 5$	$BC = 5$	$AC = 7.07$
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Conclusion (instructor-judged). A correct response says the triangle is isosceles because two sides are equal and the third is not: AB and BC are both 5, while AC is about 7.07. Three lengths with no conclusion is not an argument.

- 11.** Pool 25 m by 10 m by 2 m deep. Find the surface diagonal and the rope.

Solution: Stage one is the rectangle at the surface: $\sqrt{25^2 + 10^2} = \sqrt{625 + 100} = \sqrt{725} = 26.9258\dots$, which rounds to 26.93. Stage two adds the depth as a third leg:

$$\sqrt{25^2 + 10^2 + 2^2} = \sqrt{625 + 100 + 4} = \sqrt{729} = 27$$

surface diagonal = 26.93 m	rope = 27 m
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- 12.** Garden $(0, 0)$, $(12, 0)$, $(12, 5)$, $(0, 5)$ in metres. Find the support, the perimeter, and the total.

Solution: The support is the diagonal of the rectangle: $\sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$. The perimeter is $2(12 + 5) = 34$, and the two together come to $34 + 13$.

support = 13 m	perimeter = 34 m	total = 47 m
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Reason (instructor-judged). A correct response says half the perimeter is exactly $12 + 5 = 17$, the two legs added, while the support is the hypotenuse of that same right triangle — and a hypotenuse is always shorter than the two legs added together, so $13 < 17$. Accept any wording that says the straight route beats going along two sides.